



Unifying RNA-seq data using meta-analysis: Bioinformatics frameworks and application for plant genomics

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ABSTRACT

RNA sequencing (RNA-Seq) has transformed plant genomics by enabling high-resolution profiling of gene expression across various conditions. However, integrating RNA-Seq data from different studies is challenging due to variability in experimental designs, sequencing platforms, and data processing workflows, which limits the comparability and applicability of transcriptomic datasets. This review provides an overview of current meta-analysis approaches that address these challenges and enhance the consistency, accuracy, and interpretability of RNA-Seq data integration. We discuss methodologies such as data normalization techniques, statistical frameworks for aggregating results, and computational tools that reduce inter-study variability. We also highlight preprocessing strategies, including batch effect correction and standardized gene annotation pipelines, which facilitate reliable cross-study comparisons. We emphasize the practical significance of RNA-Seq meta-analysis in plant genomics. Meta-analysis improves the identification of consistent differentially expressed genes (DEGs), enhances functional annotation, and uncovers conserved regulatory mechanisms across plant species. These insights have applications in precision breeding, stress-response studies, and trait improvement programs. For researchers implementing meta-analysis, this review outlines key considerations, recommended practices, and available resources. We conclude by highlighting the need for standardized protocols and promoting multi-omics integration to unlock deeper insights. As transcriptomic datasets expand, meta-analysis will play a crucial role in advancing our understanding of plant biology and its application in agriculture.

1. Introduction

In recent years, the availability of transcriptome data in the field of plant genomics has increased significantly. This increase is due to advances in high-throughput sequencing technologies. As a result, researchers can now study gene expression profiles in different plant species and under different experimental conditions [1]. This wealth of data has provided valuable insights into plant biology, including stress responses, development and adaptation [2–4]. By analyzing the transcriptome, scientists can obtain a snapshot of gene activity at a specific point in time. This allows them to have a better understanding of the intricate regulatory networks and functional dynamics that govern plant life.

The advent of RNA sequencing (RNA-Seq) technology has revolutionized plant genomics by enabling comprehensive transcriptome

profiling across different developmental stages, environmental conditions and genetic backgrounds [5]. RNA-Seq provides a holistic view of gene expression by capturing the abundance and diversity of transcripts in plant cells. This high-throughput approach has enabled researchers to study complicated biological processes such as gene regulation, stress responses and metabolic pathways in unprecedented depth. For example, recent RNA-Seq meta-analyses have identified conserved stress-responsive genes in various species, such as those involved in abscisic acid signaling pathways in *Arabidopsis thaliana* under drought and salt stress [2], and key transcriptomic responses to both abiotic and biotic stress factors in rice [6]. However, these results reflect the diversity of approaches used to integrate RNA-Seq data. In studies by Shintani et al. [1] quantile normalization was used to deal with batch effects, while Rau et al. [7] favored variance-stabilizing transformations for improved consistency between datasets. This highlights the ongoing

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debate about the most effective normalization and statistical methods for RNA-Seq meta-analysis. Furthermore, while these studies have demonstrated the potential of meta-analysis in identifying conserved gene networks, they also highlight the challenges posed by dataset heterogeneity and the need for more refined integration techniques. The generation of large amounts of transcriptomic data from different experimental setups provides a solid foundation for understanding plant biology at the molecular level [8]. However, due to the complexity and volume of these data, advanced analytical strategies are often required to gain meaningful biological insights.

In addition to abiotic stress and environmental factors, the physiological effects of agricultural inputs, such as agrochemicals and biological pesticides, should also be taken into account. These agents, which are widely used in modern agriculture, can have profound effects on gene expression and influence plant health, growth and stress tolerance [9]. Incorporating these factors into RNA-Seq studies enables a more comprehensive understanding of plant biology and its responses to agricultural practices and improves the applicability of genomic data to real-world agricultural challenges [10]. In addition, the increasing focus on sustainable agriculture highlights the need to identify genes and signaling pathways that contribute to resilience and productivity under different environmental conditions.

Despite the transformative potential of RNA-Seq, the integration of data from multiple studies through meta-analysis presents both opportunities and challenges. The main advantage of meta-analysis is that it brings together the results of different studies and provides a unified perspective on gene expression patterns [11]. This approach can reveal consistent biological signals that may remain hidden when looking at individual data sets in isolation. For instance, advancements in methodologies like the integration of p-value combination techniques, such as Fisher's combined probability test and Stouffer's Z-score method, enable a more robust meta-analysis of diverse datasets [12,13]. In addition, meta-analysis has proven to be a powerful tool for identifying stress-responsive genes, uncovering conserved regulatory networks and discovering new candidates for crop improvement. However, the effectiveness of meta-analysis depends on solving several critical problems associated with the integration of RNA-Seq data. Variability in experimental design, differences in sequencing technologies, and differences in data quality can introduce significant bias and make it difficult to obtain reliable results from different studies [7]. In addition, previous meta-analyses have encountered challenges related to inadequate normalization techniques and incomplete correction for batch effects, leading to potential inaccuracies and inconsistencies in the integrated data. Recognizing these challenges is critical to improving the reliability and impact of meta-analyses in plant genomics. While integrating data from several datasets, it is essential to use effective normalization and robust methods for correction, to ensure that the combined results reflect true biological variation and not technical artefacts [14]. Furthermore, addressing variability in study designs and data quality requires the development of standardized protocols and advanced computational tools to enable more precise and reproducible meta-analysis results [15]. These advances have the potential to close gaps in current methods and maximize the utility of RNA-Seq data in crop improvement programmers.

The introduction of RNA-Seq marked a major turning point in the field of transcriptomic meta-analysis, especially in plant research. RNA-Seq offered substantially higher sensitivity, broader dynamic range, and genome-wide resolution compared to earlier microarray platforms, enabling more detailed and quantitative comparisons across diverse datasets [16]. Accordingly, analytical methods evolved in tandem with these technological advances.

Researchers began implementing formal statistical frameworks—such as fixed- and random-effects models—and combining p-values using methods like Fisher's and Stouffer's, alongside effect-size meta-analysis (e.g. log fold-change) [17]. Specialized tools including MetaDE, RankProd, and metaRNASeq emerged to apply these models to

transcriptomic data, improving reproducibility and managing inter-study variability [18].

In parallel, significant advances occurred in data preprocessing and harmonization. Approaches like batch effect correction (e.g. ComBat), quartile normalization, and re-annotation pipelines enhanced compatibility across datasets from different platforms [19].

The biological applications of transcriptomic meta-analysis also expanded considerably. It began to incorporate time-series studies, tissue-specific expression profiles, and complex multi-factor stress experiments [20]. These were complemented by downstream functional analyses such as gene ontology (GO) enrichment, transcription factor motif prediction, and co-expression network analysis—helping to contextualize differentially expressed genes (DEGs) within broader regulatory frameworks [21,22].

This review aims to provide a comprehensive overview of the methods and applications of meta-analysis in the context of RNA-Seq data for plant genomics. Current approaches to transcriptomic data integration, including normalization and statistical models, will be explored and the advantages of Obtaining data from multiple studies will be highlighted. In addition, the existing challenges and limitations associated with meta-analysis and how these issues can be overcome to improve the accuracy and utility of the integrated results will be addressed. By consolidating recent advances and exploring practical applications, such as the identification of critical pathways using multi-study data, this paper aims to advance the understanding of how meta-analysis can be used to drive progress in crop research and biotechnology and ultimately contribute to the development of improved crop varieties and sustainable agricultural practices. With this approach, the review emphasizes the crucial role of RNA-Seq meta-analysis in overcoming bottlenecks and opening up opportunities for functional genomics and precision breeding.

2. Data collection and preparation

A major challenge in meta-analyzing RNA-Seq data is the heterogeneity and variability across studies. Differences in experimental design, sequencing technologies, and data quality can introduce inconsistencies that complicate data integration [23]. For example, variations in library preparation methods, sequencing depths, and batch effects can create biases that obscure true biological signals [24]. These limitations emphasize the need for standardized protocols and sophisticated statistical methods to ensure that meta-analytic findings accurately reflect the underlying biological phenomena rather than artifacts of technical variability. To address these challenges, effective data aggregation and preprocessing are crucial steps in preparing RNA-Seq data for meta-analysis [25]. Data aggregation involves systematically collecting RNA-Seq datasets from various sources, such as public repositories like NCBI's Gene Expression Omnibus (GEO) and EMBL-EBI's ArrayExpress. Ensuring the consistently format and quality of datasets is essential for successful integration. The preprocessing phase includes several key procedures such as quality control to assess and clean raw sequencing data, read trimming to remove low-quality or adapter sequences, and alignment to reference genomes to accurately map RNA-Seq reads. Following alignment, quantification of gene expression is performed using tools like feature Counts or Kallisto to generate expression matrices that can be compared across different studies (Fig. 1).

2.1. Normalization strategies

The combination of different datasets poses challenges due to variability arising from different experimental designs, sequencing depths, library preparation protocols, and bioinformatics pipelines. This heterogeneity can obscure biological signals and lead to potentially misleading findings if not adequately addressed. A critical step in overcoming these challenges is normalization, which adjusts for technical variation to ensure comparability of expression values across

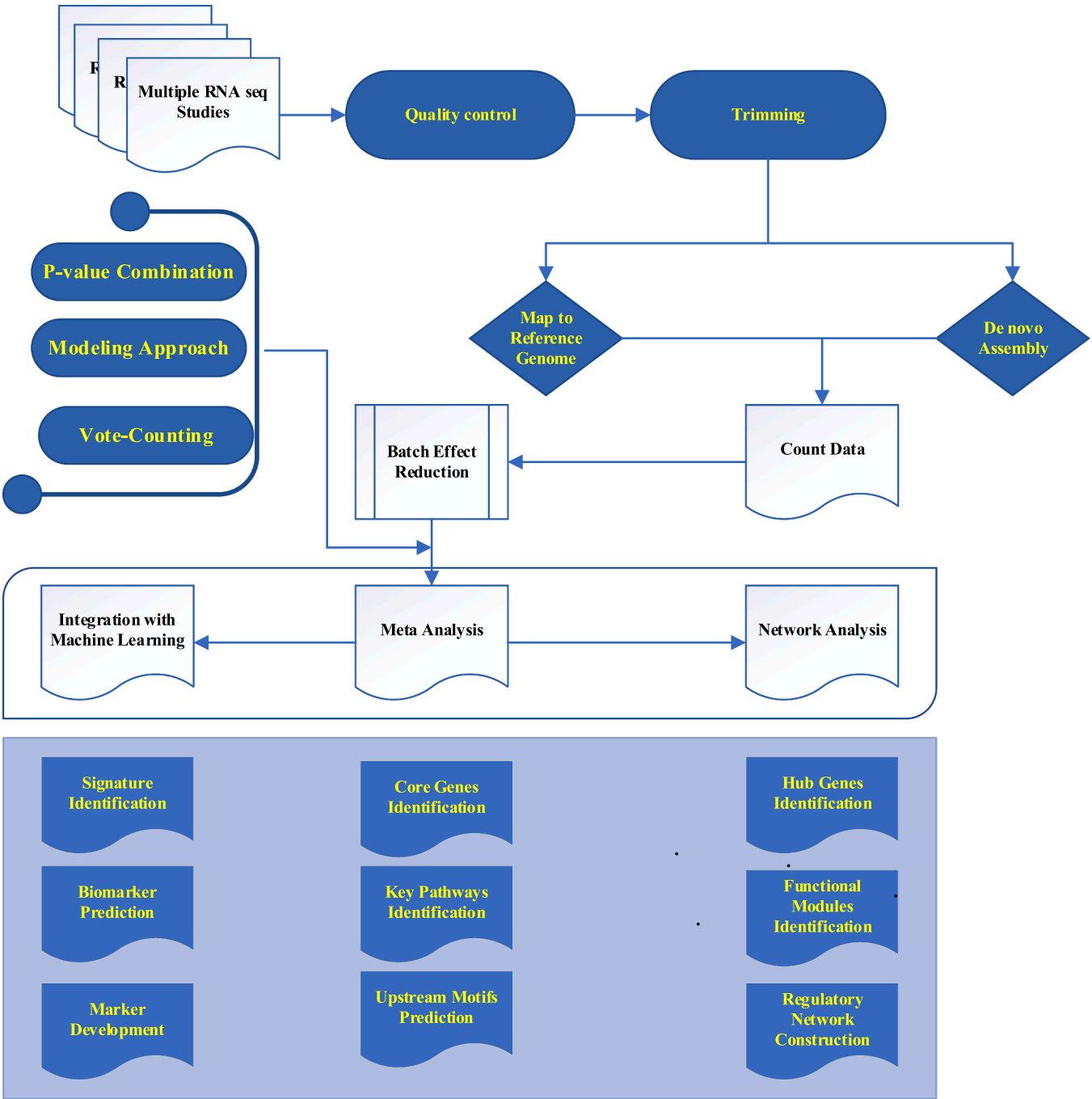


Fig. 1. Workflow and Downstream Analysis of RNA-seq Meta-Analysis. This figure illustrates the comprehensive workflow for performing meta-analysis on RNA-seq data, including key steps and downstream analyses.

samples and studies [26].

Normalization addresses discrepancies in sequencing depth and composition bias while aligning the dynamic range and distribution of expression values across datasets. Various normalization strategies, including scaling-based, distribution-based, and batch-effect correction methods, have been developed to address these issues. Each method carries distinct assumptions, strengths, and limitations that must be carefully considered within the context of meta-analysis. Moreover, normalization is crucial for ensuring the validity and reproducibility of transcriptome meta-analysis. The thoughtful selection and application of these methods—often in combination—can enhance data harmonization, facilitate robust downstream analyses, and improve the identification of consistent and biologically relevant gene expression patterns

across studies [12,27]. The following is a detailed discussion of normalization strategies commonly utilized in transcriptome meta-analysis.

3. Within-study normalization

Before integrating data from multiple RNA-Seq studies, normalization within each dataset is essential. This process accounts for technical variability and ensures comparability of gene expression values. Within-study normalization addresses systematic biases, including variations in sequencing depth, RNA composition, and library size, which can obscure biological signals. Several well-established methods are commonly employed in transcriptome analysis to achieve this objective:

- **DESeq2 Median-of-Ratios Method:** This method calculates size factors for each sample by determining the geometric mean of counts for each gene across all samples. Each gene count is divided by this mean, and the median of the resulting ratios serves as the scaling factor. This approach normalizes for sequencing depth and compositional biases, particularly in datasets where a minority of genes show significant expression changes [28].
- **edgeR Trimmed Mean of M-values (TMM):** The TMM method estimates normalization factors by calculating log-fold changes and absolute expression differences between samples. It trims extreme values—typically associated with highly expressed genes or outliers—to provide a robust adjustment. TMM is suited for datasets with varying RNA composition or when the assumption of global expression stability is valid [29].
- **Quantile Normalization:** This method transforms the distribution of gene expression values to ensure that all samples have identical statistical properties. By aligning the quantiles across samples, it mitigates technical variability in distribution shapes. However, quantile normalization assumes that the overall expression distribution should be consistent across samples, which may not hold true in all RNA-Seq contexts [30](Table 1).

4. Cross-study normalization and harmonization

After addressing biases within individual studies, it is essential to apply normalization to account for technical and biological differences across multiple studies. Cross-study normalization minimizes inter-experiment variability due to differences in sequencing platforms, library preparation protocols, and laboratory-specific conditions. This process enhances dataset comparability, increasing the robustness and reproducibility of meta-analytic results.

- **ComBat (Empirical Bayes Correction):** Developed for microarray analysis and now widely used in RNA-Seq workflows, ComBat adjusts for known batch effects by modeling and eliminating batch-specific mean and variance shifts [30]. It preserves biological variability while correcting for unwanted technical noise, making it effective for harmonizing datasets from diverse sources.
- **Z-score Standardization:** This method transforms expression values for each gene independently by centering them around a mean of zero and scaling to unit variance. Z-score transformation facilitates direct comparisons of expression profiles across datasets regardless of their original scales or units [31].
- **Rank-Based Normalization (e.g., RankProd):** This approach uses the rank order of gene expression within each dataset rather than absolute expression values. By focusing on relative expression patterns, rank-based normalization is more resilient to platform-specific biases and effective in identifying consistently differentially expressed genes across heterogeneous datasets [32] (Table 1).

5. Gene reannotation and feature matching

A critical challenge in transcriptome meta-analysis is the

inconsistency of gene identifiers and annotations across studies. Variations in reference genomes, annotation releases, or naming conventions can impede direct comparison and integration. Gene reannotation involves mapping gene identifiers from individual datasets to a common reference, such as Ensembl or RefSeq. This harmonization ensures that orthologous or homologous genes are accurately matched, enabling valid aggregation and comparison of gene-level results across studies.

6. Transformation techniques for count data

RNA-Seq data are inherently count-based, exhibiting overdispersion and a strong mean-variance relationship that complicates downstream statistical analyses. Transformation techniques stabilize variance and approximate homoscedasticity.

- **Variance Stabilizing Transformation (VST):** Implemented within the DESeq2 framework, VST reduces the dependency of variance on mean expression levels [33]. It is useful for preparing data for visualization, clustering, or correlation analyses by creating a more uniform variance across the expression range.
- **Regularized Logarithm (rlog):** rlog available in DESeq2, rlog applies a log transformation in conjunction with the shrinkage of dispersion estimates. This method stabilizes variance while preserving the data structure, making it suitable for datasets with few replicates or high variability [34].

7. Statistical approaches

In the context of meta-analysis, statistical models play a critical role in synthesizing results from multiple studies to generate more generalized conclusions. The two primary types of models used are fixed-effects and random-effects models, each serving distinct purposes depending on the nature of the data and the objectives of the analysis.

8. Fixed-effects models

Fixed-effects models originated in the early 20th century, assume that the effect size remains constant across all studies and any observed variability is solely due to sampling error [35]. This approach is appropriate when the studies included in the meta-analysis are highly similar in terms of their design, populations, and measurements, aiming to estimate a single effect size that reflects this commonality. Fixed-effects models are best suited for scenarios where homogeneity between studies is guaranteed, such as analyses involving standardized protocols or controlled experimental conditions. However, this model can be overly restrictive if there is significant variation between studies, as it does not account for differences that may arise from heterogeneity in study conditions or populations [36].

9. Random-effects models

Random-effects models introduced in the latter half of the 20th century, address the limitations of fixed-effects models by accounting for

Table 1
Advantages and disadvantages of normalization methods in RNA-seq meta-analysis.

Normalization method	Category	Advantages	Disadvantages
DESeq2 Median-of-Ratios	Within-study	Robust to outliers; handles library size differences	Assumes most genes not differentially expressed
edgeR TMM	Within-study	Handles composition bias; effective for unbalanced data	May underperform with global expression changes
Quantile Normalization	Within-study	Simple; makes distributions identical	May remove biological differences
ComBat	Cross-study	Corrects batch effects; preserves signal	Needs batch labels; overfitting risk in small samples
Z-score Standardization	Cross-study	Easy to implement; useful for meta-analysis	Reduces absolute signal intensity
Rank-Based Normalization	Cross-study	Minimizes platform bias; robust to outliers	Loses quantitative information
Gene Reannotation	Feature matching	Enhances cross-study consistency	Prone to mapping errors if annotations are inconsistent
VST	Data transformation	Stabilizes variance; good for clustering	May distort low-count data
Rlog	Data transformation	Regularizes variance; useful in small datasets	Computationally expensive

variability across studies. These models posit that the effect sizes may vary from study to study due to differences in study populations, methodologies, and other factors, in addition to sampling error. Random-effects models estimate the average effect size while considering this between-study variability, offering a more flexible approach that can better handle the heterogeneity often encountered in meta-analyses [37]. They are particularly applicable to RNA-seq studies, where substantial variability in experimental protocols, sequencing platforms, and sample preparations is common.

Fixed-effects and random-effects models have distinct strengths and limitations when utilized in meta-analysis, particularly in fields such as RNA-seq (Table 1). Fixed-effects models are valued for their simplicity and clarity. They provide a straightforward estimate of the effect size by averaging results from studies assumed to have homogeneous designs, populations, and methods [38]. This model can yield precise estimates with narrow confidence intervals when the studies are indeed similar. However, this approach assumes that all studies are assessing the same underlying effect size, which may not be accurate if there are significant differences between studies, leading to potential bias [39].

On the other hand, random-effects models address between-study variability by assuming that effect sizes may vary across studies due to differences in populations, methodologies, or other factors, in addition to sampling error [40]. By incorporating both within-study and between-study variability, these models provide a more generalized estimate of the effect size, making them particularly suitable for RNA-seq meta-analyses where heterogeneity in experimental conditions is common. However, challenges such as accurately estimating between-study variance and their increased complexity compared to fixed-effects models must be considered [27]. Recent advancements, including Bayesian hierarchical models and machine-learning frameworks, have further enhanced the robustness and applicability of random-effects models for handling the complexity of RNA-seq datasets with high dimensionality and missing data [41]. While fixed-effects models remain a valuable choice for highly homogeneous datasets, random-effects models and their modern extensions, such as Bayesian models, are indispensable for addressing the challenges posed by heterogeneity in RNA-seq studies [42]. By understanding the evolution and scope of these methodologies, researchers can tailor their analyses to align with the characteristics of their datasets and the objectives of their studies.

10. Fisher’s combined probability test

In the meta-analysis of RNA-seq data, integrating significance across multiple studies often involves combining p-values to draw more robust conclusions about gene expression [4,12]. Two commonly used methods for this purpose are Fisher’s combined probability test and Stouffer’s Z-score method[12]. These methods offer different approaches to aggregating p-values, each with its advantages.

Fisher’s combined probability test is a widely used method for meta-analyzing p-values from different studies [43]. This test combines the p-values into a single test statistic, which is then used to assess overall significance.

11. Stouffer’s Z-score method

Stouffer’s Z-score method combines Z-scores derived from p-values, providing an alternative approach [44]. This method standardizes the effect sizes and then combines them, making it particularly helpful when dealing with studies that have different sample sizes. It proves particularly useful when combining studies with different sample sizes and can provide more nuanced insights when integrating results across studies.

Fisher’s combined probability test and Stouffer’s Z-score method are widely used techniques for integrating p-values from multiple studies in meta-analysis, each with its own strengths and limitations. Fisher’s combined probability test is valued for its simplicity and ease of

implementation. It combines p-values into a single Chi-Square (χ^2) statistic, which is straightforward to calculate and interpret. This method does not require assumptions about the distribution of effect sizes or the direction of effects across studies, making it a flexible option for integrating results from studies with different hypotheses. Additionally, Fisher’s test is effective when individual studies have small sample sizes or when effect sizes are not directly comparable (Table 2). However, it is sensitive to extremely small p-values, which can overshadow results from studies with more moderate p-values [13]. The method also assumes that p-values are independent, an assumption that may not hold in cases where studies are correlated or share similar data sources. Furthermore, Fisher’s test does not account for the size or precision of individual studies, potentially leading to less accurate results when studies vary significantly in sample size or statistical power.

Stouffer’s Z-score method, on the other hand, incorporates study weights based on sample size or other measures of study precision, allowing it to account for variations in the quality and precision of the study. This method is robust for analyzing variations in sample size, effectively aggregating results from studies with different statistical power. By converting p-values to Z-scores and combining them, Stouffer’s method is less affected by extremely small p-values compared to Fisher’s test, providing a more balanced integration of significance

Table 2
Comparison of meta-analysis methods. This table provides an overview of the main methods used for meta-analysis of RNA-seq data, including their advantages, disadvantages, and limitations.

Method	Advantages	Disadvantages	Limitations
Vote-Counting	- Simple and easy to implement. - Provides a quick overview of consistency	- Does not account for the magnitude or direction of effects. - Can be influenced by publication bias. - Doesn't handle variability among studies.	- Lacks statistical rigor. - May overlook small effects. - Doesn't consider study quality.
Stouffer’s Z-Score Method	- Combines p-values across studies using z-scores, which accounts for sample size. - Provides a weighted summary of significance.	- Assumes that the effect sizes are normally distributed. - Requires the same direction of effect for meaningful combination.	- Sensitive to extreme p-values. - Assumes studies are of equal quality.
Fisher’s Combined Probability Test	- Simple and widely used. - Does not require effect size estimates, just p-values.	- Assumes independence of studies. - Does not account for variability in effect sizes. - Can be overly sensitive to very small p-values.	- Less effective with heterogeneous studies. - Limited in addressing study quality and variability.
Random-Effects Model	- Accounts for between-study variability (heterogeneity). - Provides a more generalizable estimate of the effect size.	- More complex and requires accurate estimation of heterogeneity. - Can be less precise with few studies.	- Can be influenced by large variability between studies. - Requires careful interpretation of random effects.
Fixed-Effects Model	- Assumes that all studies estimate the same underlying effect size. - Simpler to implement when studies are similar.	- Assumes homogeneity among studies, which may not hold true. - Not suitable if studies are highly heterogeneous.	- Less generalizable if there is significant variability among studies. - May give misleading results if assumptions are violated.

levels. However, Stouffer's method is more complex to compute, requiring the calculation of Z-scores and appropriate weights, which adds to its computational demands. Additionally, it assumes that Z-scores are normally distributed, which may not always be the case, potentially impacting the accuracy of the results [13]. The method is also dependent on the selection of weights; inappropriate weighting can skew the combined results (Table 2).

12. Computational tools and software

In the domain of meta-analysis methods for RNA-Seq data, it is crucial to acknowledge and rectify the limitations of previous research in order to advance the field and enhance the accuracy and reliability of results. Each method and package discussed, has its own merits and caters to various requirements, but recognizing and addressing their shortcomings can significantly improve the overall quality of studies.

metaRNASeq is well-known for its simplicity and ease of implementation, making it advantageous for users seeking straightforward solutions. However, its sole-reliance on p-values can be a significant constraint, especially when dealing with heterogeneity across studies. Previous research often struggled when integrating diverse datasets, where p-values may have failed to capture the full range of effect sizes and study variability. The limited ability of this method to handle heterogeneity emphasizes the need for more comprehensive approaches that take both p-values and effect sizes into consideration [45]. Despite this limitation, metaRNASeq remains a widely used tool due to its effectiveness in integrating data from studies with similar experimental designs.

metaseqR2 offers a comprehensive approach with its extensive normalization and diverse range of statistical tests. While this package addresses the need for careful preprocessing and multiple testing methodologies, its complexity and the requirement for extensive preprocessing are noteworthy limitations. Previous studies have shown that inadequate preprocessing can result in biased results or the oversight of significant findings [46]. Therefore, it is essential to address the intricacies of preprocessing and ensure accessibility for users who are less familiar with RNA-Seq, ultimately enhancing the outcomes of meta-analyses.

DExMA is known for its capability to handle both fixed- and random-effects models, as well as missing data. However, the fixed-effects model may oversimplify complex data, and effectively handling missing data can be challenging [47]. Previous research often faced obstacles when dealing with data heterogeneity and missing values, which compromised the stability of meta-analysis results. Improved methodologies for handling missing data and adopting a balanced approach to fixed- and random-effects models could enhance the reliability of conclusions derived from meta-analyses.

MetaSeq is highly effective in the normalization process and can integrate data from multiple sources, making it an essential component for meta-analysis [48]. However, it is important to note that the log-transformation used may not be suitable for all datasets, and precise normalization is crucial to avoid skewed results. MetaSeq's robust normalization capabilities are valuable, but users must ensure that the chosen methods align with the characteristics of their specific datasets.

GeneMeta utilizes empirical Bayes methods to generate robust estimates for high-throughput data. However, it is important to consider the limitations of this approach, particularly its computational intensity and effectiveness with small sample sizes [49]. Despite its computational intensity, GeneMeta is widely used for large datasets, where its accuracy and robustness are critical for meta-analysis.

MetaDE employs a comprehensive approach to meta-analysis by combining effect sizes and p-values [50]. However, it may not address data heterogeneity as effectively as more advanced methods. Previous researches have suggested that combining effect sizes and p-values can offer a more nuanced perspective, but addressing the issue of heterogeneity is still important. Enhancing methods to adequately account for

variability between studies can greatly improve the accuracy of meta-analysis. MetaDE's strength lies in its ability to incorporate both p-values and effect sizes, making it a versatile tool for RNA-Seq meta-analysis. However, it requires careful consideration of heterogeneity to maximize its utility.

MetaGen provides valuable insights by incorporating gene set enrichment analysis alongside fixed- and random-effects models [51]. However, it is important to recognize that this approach may require extensive preprocessing and may not be suitable for all study designs. MetaGen's integration of gene set enrichment makes it a powerful tool for functional analysis, although the extensive preprocessing required may limit its accessibility for some users.

MetaNorm prioritizes normalization and integrates it with meta-analysis, but it is crucial to find a balance between normalization and robust meta-analysis techniques [52]. Focusing solely on normalization could limit MetaNorm's capabilities. Previous research has emphasized the importance of proper normalization, but comprehensive meta-analysis methods are equally vital. Striking a balance between these aspects can enhance the overall analytical process. MetaNorm is an essential tool for normalization, but for comprehensive meta-analysis, additional methods for handling heterogeneity and effect sizes must be considered (Table 3).

metafor offers a flexible framework that integrates functional data with various meta-analysis methods [53]. However, it is worth noting that this framework may be less specialized for RNA-Seq and requires a strong understanding of functional data integration (Table 3). Previous studies have acknowledged the complexity of integrating functional data and the necessity for specialized approaches. Improving the framework's usability and tailoring the methods specifically for RNA-Seq data can significantly enhance its effectiveness. metafor's flexibility allows it to be adapted for various types of data, but it may require a deeper understanding of functional data integration to fully leverage its potential.

SMAGEXP (Galaxy) seamlessly integrates with the Galaxy platform and offers a wide range of meta-analysis methods [54]. Nevertheless, the reliance on the Galaxy environment and the need for users to be familiar with the platform may present limitations. Previous studies have recognized the benefits of integrating with platforms like Galaxy, but they have also acknowledged the associated drawbacks. Expanding the accessibility and functionality beyond specific platforms can broaden the usability of meta-analysis tools. Moreover, applying different computational tools and software to the same dataset and comparing the results can provide valuable insights into the consistency and robustness of the findings. Such comparative analyses can reveal the strengths and weaknesses of each tool, helping researchers choose the most appropriate methods for their specific dataset and improving the overall quality of meta-analytic outcomes [55]. Addressing these constraints in future research endeavors can enhance the precision, efficiency, and practicality of meta-analysis methods for RNA-Seq data, thus enabling more reliable and actionable scientific insights. SMAGEXP's integration with Galaxy enhances its utility but requires user familiarity with the platform. Expanding its accessibility could make it a more universally applicable tool.

Several meta-analysis tools and packages have been developed to help with the integration and interpretation of RNA-Seq data. These programs, including metaRNASeq, metaseqR2 and DExMA, offer different techniques and methods for managing RNA-Seq data. Table 3 compares different meta-analysis packages and describes their strengths, limitations and data processing capabilities to help researchers select the best tool for their individual meta-analysis needs.

13. Examples for different meta-analysis procedures

RNA-Seq meta-analysis uses different data analysis methods tailored to the research question, the characteristics of the data set and the degree of variability between the studies to be integrated. The technical

Table 3
Comparison of meta-analysis packages for RNA-seq data.

Package	Algorithms/Methods	Strengths	Limitations	Data Handling	Meta-analysis models
metaRNASeq	Inverse Normal Method - Fisher's Combined Probability Test	- Simple and easy to implement. - Good documentation	Limited to p-values. - May not handle heterogeneity well.	P-values only; no effect size consideration.	Limited to p-value combination methods.
metaseqR2	- Normalization Algorithms - Statistical Testing Methods	- Comprehensive normalization. - Various statistical tests included.	Requires extensive preprocessing. - Complex for users unfamiliar with RNA-Seq	Comprehensive preprocessing required.	Includes various methods but not specifically meta-analysis focused
DExMA	Fixed-Effects Meta-Analysis - Random-Effects Meta-Analysis - Handling Missing Data	- Handles both fixed and random effects. - Deals with missing data.	- Fixed-effects model may be too simplistic. - Handling missing data can be challenging.	Handles both fixed and random effects; can manage missing data.	Provides fixed and random effects models for meta-analysis.
MetaSeq	Log-Transformation - Normalization and Variance Stabilization - Meta-Analysis Models	- Robust normalization. - Integrates data from multiple sources.	- Log-transformation may not suit all datasets. - Requires precise normalization.	Integrates various sources; robust normalization	Uses meta-analysis models to combine data from multiple studies.
GeneMeta	- Meta-Analysis with Empirical Bayes Methods - Statistical Tests for Differential Expression	Uses empirical Bayes for robust estimates. - Tailored for high-throughput data	- Computationally intensive. - Less effective with small sample sizes.	Empirical Bayes for combining data; robust for high-throughput data	Focuses on empirical Bayes methods for meta-analysis.
MetaDE	Fixed-Effects Meta-Analysis - Random-Effects Meta-Analysis - Effect Size Combination	- Offers both fixed and random effects models. - Considers effect sizes for meta-analysis.	- May not handle data heterogeneity as effectively as some advanced methods.	Effect sizes and p-values	Combines effect sizes and p-values for meta-analysis.
MetaGen	Gene Set Enrichment Analysis - Fixed and Random Effects Models	- Includes gene set enrichment analysis. - Provides both fixed and random effects models.	May require extensive preprocessing. - Gene set analysis may not fit all study designs.	Gene expression data and associated meta-analysis.	Provides fixed and random effects models with enrichment analysis.
MetaNorm	- Normalization Procedures - Meta-Analysis of Normalized Data	Specializes in normalization. - Integrates normalization with meta-analysis.	Focuses primarily on normalization; may not provide extensive meta-analysis methods.	Normalized data from multiple studies.	Meta-analysis using normalized data.
metafor	- Fixed-Effects Meta-Analysis - Random-Effects Meta-Analysis - Fisher's Combined Probability Test - Stouffer's Z-Score Method	- Integrates with functional data and offers a flexible meta-analysis framework.	- May be less specialized for RNA-Seq compared to other packages. - Requires understanding of functional data integration	Functional and meta-analysis data.	Flexible meta-analysis framework integrating functional data.
SMAGEXP (Galaxy)	- Fixed-Effects Meta-Analysis - Random-Effects Meta-Analysis - Fisher's Combined Probability Test - Stouffer's Z-Score Method	- Accessible via Galaxy platform. - Integrates well with other Galaxy tools for analysis.	- May require familiarity with Galaxy platform. - Limited to Galaxy environment constraints.	Data integrated with Galaxy tools.	Statistical meta-analysis integrated with Galaxy tools.

discussion of different meta-analysis algorithms provided in appendix 1. However, we discussed the examples of different analytical approaches and their applications in plant genomics:

1. Fixed-Effects Models

Fixed-effects models are often used when the studies included in a meta-analysis are expected to have relatively comparable designs, settings and sample populations. For example, when studying *Arabidopsis thaliana* gene expression under drought stress, a fixed-effects model can be used to summarize the results of many controlled trials with minimal variance in experimental design. The fixed-effects models was used to integrate drought-responsive gene expression data across controlled laboratory studies of *Arabidopsis* in a study by Rau et al. (2014), leading to the identification of conserved drought-related genes [7]. Similarly, Tamura and Bono (2022) employed fixed-effects models in a meta-analysis of hypoxia-responsive genes in *Arabidopsis* and rice, effectively identifying a core set of

hypoxia-inducible genes despite slight variations in experimental protocols [56]. This method predicts a single, pooled effect size and is particularly well suited to finding consistent gene expression patterns when the studies have similar general conditions, leading to strong results for specific research questions [3].

2. Random-Effects Models

Random effects models are better suited for studies with significant variability due to variations in experimental designs, demographics or sequencing platforms. For example, when integrating RNA-Seq data from different plant species such as maize, rice and wheat under drought stress conditions, random effects models take into account the heterogeneity of the study. Benny et al. (2019) applied random-effects models to integrate drought-stress datasets across diverse crop species, revealing both conserved and species-specific transcriptional responses [57]. Another example is the work by Balti et al. (2020), which used random-effects models to identify genes associated with abiotic stress in various plants,

demonstrating their applicability in handling cross-species variability and diverse study designs [58]. These models provide a more generalizable estimate of effect size by accounting for both within-study and between-study variability. This accounts for heterogeneity and increases the applicability of the results to different data sets [57].

3. Bayesian Hierarchical Models

Bayesian hierarchical models provide a versatile strategy for dealing with high-dimensional data sets, missing data or substantial heterogeneity. This approach is particularly useful in studies with a limited number of samples or surveys, as it allows past information to be included in the analysis. For example, Bayesian models have been used in meta-analyses of gene expression under drought and salt stress to refine predictions and reveal intricate regulatory networks that are conserved across species and adapt to specific environmental conditions. For instance, Cohen et al. (2019) utilized Bayesian hierarchical models to integrate biotic and abiotic stress response datasets in rice, uncovering shared and unique regulatory pathways [6]. Another study by Shaar-Moshe et al. (2015) employed this approach to identify conserved drought-adaptive genes across species, incorporating prior knowledge to refine their predictions [59]. This strategy has been particularly successful in discovering gene regulatory networks that are conserved across species but respond differently to environmental stress [42].

4. Combining p-values and Effect Sizes

Another important strategy in RNA-Seq meta-analysis is to combine p-values from numerous studies to obtain more meaningful results. Methods such as Fisher's combined probability test and Stouffer's Z-score approach combine p-values or Z-scores from different studies to obtain an overall estimate of statistical significance. These approaches are particularly effective when it comes to combining data from studies with different sample sizes or estimating the size of direct effects. For example, a meta-analysis of RNA-Seq research on biotic stress responses in plants could use the Fisher technique to synthesize data that provide insights into conserved changes in gene expression across species [12]. The Fisher's combined probability test was used by Hu et al. (2024) in a meta-analysis of drought-responsive genes in *Oryza sativa*, uncovering pathways related to abscisic acid signaling [60]. Similarly, Pedersen et al. (2012) applied Stouffer's Z-score method to analyze gene expression changes in plants subjected to heavy metal stress, revealing conserved detoxification mechanisms across species [44].

5. Integration of Multi-Omics Data

Meta-analysis can combine transcriptome data with data from other omics platforms such as proteomics and metabolomics to provide a more complete picture of biological processes. Multi-omics data integration approaches, usually in conjunction with machine learning methods, are increasingly being used to uncover intricate gene regulatory networks and pathways. For example, combining RNA-Seq data with metabolomics data has shown how changes in gene expression correspond to metabolic adaptations under stress, improving our knowledge of plant resilience and adaptability. Wang et al. integrated transcriptomics with proteomics to explore pathogen resistance pathways in maize, leading to the discovery of novel regulatory proteins [20]. Similarly, Miransari demonstrated the utility of combining transcriptomic and metabolomic data in arbuscular mycorrhizal fungi-plant interactions, revealing enhanced phosphorus uptake mechanisms under drought stress [61]. These integrative techniques promote systems-level knowledge by linking molecular interactions across biological levels [20].

This section attempts to illustrate the variety of data processing methods used in RNA-Seq meta-analysis using concrete examples and highlighting their contextual use. These examples not only demonstrate the approaches but also emphasize their importance in addressing unique difficulties in plant genomics, such as heterogeneity, variability and data integration.

14. Interpreting meta-analysis results

Interpreting the results of meta-analyses in RNA sequencing requires a thorough understanding of both the statistical methods and the biological background, especially in the field of plant genomics. Summarizing statistics such as effect sizes and p-values are crucial for determining the extent and significance of differential gene expression in plants [24]. Effect sizes, such as fold changes or log2 ratios, provide important information about the biological significance of changes in gene expression, particularly in response to abiotic and biotic stimuli, developmental stages and adaptive traits. These measurements provide important information on how specific genes contribute to plant resilience and development under a variety of environmental situations and allow researchers to identify key markers of plant improvement and adaptation to stress. It is crucial to understand both the effect sizes and the confidence intervals to determine not only the statistical significance but also the biological significance of the results of functional plant genomics [62].

Another important part of the interpretation of meta-analyses is the assessment of heterogeneity. I^2 -statistics and Q-tests determine whether the variability in effect sizes is related to biological differences between studies, e.g. changes in plant species, experimental conditions or environmental stressors, rather than random sampling error. High heterogeneity could be due to different experimental methods or tissue types, which affects the consistency and reliability of results from different studies. Therefore, methods such as subgroup analysis and meta-regression are crucial to account for different sources of variance and improve the robustness of plant genomics results [63]. Consistency between studies is crucial for the validation of results in plant research. Researchers can determine whether conclusions are consistent by comparing the results of individual studies with those of the meta-analysis, or whether additional research is needed to understand the discrepancies between studies, especially if they involve different plant species or stress conditions [64].

Consistency between studies is crucial for the validation of results in plant research. By comparing the results of individual studies with those of the meta-analysis, researchers can determine whether the conclusions are consistent or whether further investigation is needed to understand the discrepancies between studies, especially when they involve different plant species or stress conditions [65]. Consistent results between studies strengthen the reliability of the meta-analysis results, while placing them in the plant biological context ensures that the conclusions reflect plant-specific mechanisms such as gene regulatory networks and metabolic pathways.

To understand the effects of differentially expressed genes or metabolic pathways, they must be interpreted within a larger biological framework. The aim is to determine whether the genes found are related to known biological processes such as stress responses, nutrient uptake or developmental programs that are important for the growth and adaptability of plants. Integrating these discoveries with current information on plant genomes ensures that they make a meaningful contribution to our understanding of plant resilience, adaptability and growth [66].

Functional enrichment analyses, such as pathway and Gene Ontology (GO) word analyses, shed light on the molecular processes responsible for the reported changes in gene expression in plants. These methods help in the identification of over-represented pathways or biological processes among the differentially expressed genes, focusing on pathways involved in stress response, growth control, food acquisition and reproductive development [67,68].

Dealing with possible biases is crucial for a correct interpretation. The results of meta-analyses can be influenced by publication bias. These occur when studies with significant results are more likely to be published. Tools such as funnel plots and statistical tests for publication bias can help to identify such problems and ensure the authenticity of the results [69]. In addition, sensitivity analyses are needed to check the

robustness of results and ensure consistency of conclusions in case of different analytical choices or exclusion of certain research. Addressing potential biases in data collection, research design and interpretation can help to ensure that the results of the meta-analysis accurately reflect biological variance and are not influenced by methodological artefacts. Finally, it is important to analyse the significance of these results in the context of plant biology. This includes recognizing limitations such as methodological boundaries or sources of heterogeneity that may influence the interpretation of results [70,71].

In summary, interpreting the results of meta-analyses at RNA-Seq is a multi-faceted process that involves understanding statistical measures, assessing heterogeneity, ensuring consistency and integrating the results into the context of plant genomics. By focusing on the biological relevance of effect sizes and integrating statistical findings into the existing knowledge of plant genomics, meta-analysis provides a powerful framework for uncovering gene regulatory networks and molecular mechanisms that contribute to plant stress tolerance and adaptation [72]. By addressing bias and assessing the robustness of results, we can ensure that conclusions are both reliable and meaningful to advance plant biology and biotechnology.

15. Transforming transcriptomic meta-analysis using artificial intelligence (AI) approach

Artificial intelligence (AI) is increasingly transforming transcriptomic meta-analysis, offering powerful tools to address the challenges posed by data heterogeneity, platform variability, and complex biological interactions in RNA-Seq studies [73,74]. Traditional meta-analytical approaches—such as p-value combination methods (e.g., Fisher's or Stouffer's) and effect size-based models—have laid the groundwork for integrative analysis but often struggle to capture nonlinear patterns and subtle gene expression dynamics across studies. As the volume and diversity of RNA-Seq datasets grow, AI-based techniques provide more adaptive and scalable solutions for extracting biologically meaningful insights.

One of the most impactful developments is the use of deep learning models—including convolutional neural networks (CNNs), autoencoders, and recurrent neural networks (RNNs)—which are capable of learning complex, hierarchical features from high-dimensional gene expression data without the need for extensive manual preprocessing [75]. Autoencoders, for example, compress transcriptomic datasets into lower-dimensional representations that preserve key biological signals, improving tasks such as sample clustering, dimensionality reduction, and classification. These models not only enhance noise tolerance but also facilitate more robust pattern recognition across datasets [76]. This approach has been successfully harnessed to integrative analysis of single stress transcriptome data [77] and modeling responding signatures to multifactorial stress combinations in plants [78].

Transfer learning further improves generalization, particularly in plant transcriptomics where labeled datasets may be limited for non-model species [79]. By leveraging pre-trained models on well-annotated organisms such as *Arabidopsis*, researchers can apply learned representations to crop species like wheat or barley, accelerating the identification of plant specialized metabolism genes [80]. AI also contributes to improved data harmonization. Traditional methods like ComBat and quantile normalization are useful but may fall short in correcting nonlinear batch effects. In contrast, models like generative adversarial networks (GANs) and variational autoencoders (VAEs) can learn more flexible mappings to align datasets across platforms or conditions, preserving biological variation while reducing technical noise—an essential step for reliable meta-analysis [81].

Beyond data integration, AI enhances downstream biological interpretation. Graph neural networks (GNNs) and attention mechanisms have been used to infer gene regulatory networks, predict transcription factor activity, and identify key regulatory hubs across diverse conditions. These capabilities are especially valuable in the meta-analytic

context, where combining studies improves the resolution and robustness of regulatory insights [82]. Bayesian deep learning methods offer an added advantage by incorporating uncertainty estimates into predictions. This is particularly useful for noisy or imbalanced datasets, enabling researchers to assess confidence in model outputs and prioritize results for experimental validation. Probabilistic predictions also help mitigate over fitting, making the models more reliable for practical applications [83].

Despite their advantages, AI methods also face challenges. They require large, high-quality datasets and careful validation. Interpretability remains a concern, especially for complex models where the biological rationale behind predictions is not always clear. To address these issues, there is a growing emphasis on explainable AI, standardized preprocessing pipelines, open-source model sharing, and benchmarking across datasets.

16. Multi-omics data integration with RNA-seq meta-analysis

RNA-Seq meta-analysis is a powerful tool in plant systems biology, enabling the integration of transcriptional data from various studies, genotypes, developmental stages, and environmental conditions. By aggregating RNA-Seq datasets, this method enhances statistical power, improves identification of DEGs, and reveals gene expression signatures linked to specific biological processes or stress responses [84]. However, transcriptomic data provide only a partial view of the regulatory networks governing plant physiology. Gene expression is influenced by regulatory events at genomic, epigenomic, translational, and metabolic levels. Therefore, integrating multi-omics data is essential for a comprehensive understanding of plant function and trait regulation.

Integrating genomic data—including SNPs, indels, CNVs, and structural rearrangements—with RNA-Seq meta-analysis allows researchers to identify regulatory variants affecting gene expression [85]. By combining differentially expressed genes (DEGs) from meta-analysis with genome-wide association studies (GWAS) or expression quantitative trait loci (eQTL) mapping, researchers can link transcriptional patterns to genetic loci. For example, in maize and rice, this approach has identified regulatory elements related to drought tolerance, nitrogen use efficiency, and grain quality [86]. This method enhances the biological interpretation of RNA-Seq data and supports marker-assisted selection in crop breeding.

Epigenetic modifications—such as DNA methylation, histone tail modifications (e.g., H3K4me3, H3K27me3), and changes in chromatin accessibility—are crucial for regulating gene expression in response to environmental stimuli. Plants rely on epigenetic reprogramming to adapt to abiotic and biotic stresses. By integrating epigenomic data with RNA-Seq meta-analysis, researchers can identify differentially expressed genes (DEGs) regulated by epigenetic mechanisms and assess the impact of stress-induced chromatin remodeling on transcription [87]. Studies have shown that stress-responsive genes often exhibit dynamic patterns of DNA methylation or histone modifications that correlate with expression changes during drought, salinity, or pathogen exposure [88]. Additionally, integrating epigenomic data helps distinguish between transient and stable expression changes, offering insights into stress memory mechanisms and transgenerational epigenetic inheritance in plants.

RNA-Seq meta-analysis provides insights into transcript abundance, but this does not always align with protein levels due to post-transcriptional and translational regulation. In plants, where responses rely on regulated protein activities, proteomic data are crucial for validating transcriptomic findings. Quantitative proteomics, especially mass spectrometry-based techniques, allows for the identification and quantification of proteins linked to upregulated or downregulated transcripts. It also reveals post-translational modifications like phosphorylation, ubiquitination, and glycosylation, which are vital for protein stability and function. In crops, integrative transcriptome-proteome analyses have identified signaling proteins and enzymes

activated under specific stress conditions, improving our understanding of functional response pathways and identifying protein-level biomarkers for breeding [89,90].

In conclusion, integrating multi-omics data with RNA-Seq meta-analysis advances our understanding of plant biology at a systems level. RNA-Seq meta-analysis identifies consistent transcriptional patterns, while multi-omics integration offers functional depth and regulatory context. Together, these methodologies reconstruct regulatory networks, enhance gene function prediction, and inform breeding strategies. As multi-omics technologies become more accessible and computational pipelines standardized, this approach will drive innovation in plant functional genomics and crop improvement.

17. Applications of RNA seq meta-analysis in plant genomics

17.1. Key gene identification and regulatory networks inferring

RNA-Seq meta-analysis has become an invaluable tool for identifying key genes in plants. It is particularly useful because it can integrate data from diverse studies and conditions, improving the identification and validation of important genes involved in various biological processes such as stress responses, development, and disease resistance. Combining datasets from multiple RNA-Seq studies significantly increases statistical power, improving the sensitivity and specificity of gene identification [7]. By integrating results from different experiments, researchers can detect genes that consistently show differential expression across various conditions, species, or developmental stages. For example, meta-analyses have been used to identify key genes involved in plant responses to abiotic stressors such as drought, salinity, and heavy metals [91,92]. This methodology also facilitates the selection of traits critical for enhanced stress physiology, particularly in the context of gene-environment interactions and microbial associations. For instance, recent RNA-Seq meta-analyses have highlighted how plant-microbe interactions, such as symbiotic relationships with rhizobia and mycorrhizal fungi, contribute to improved nutrient uptake and resilience to drought and salinity stress. A notable example involves studies showing that arbuscular mycorrhizal fungi enhance phosphorus and water uptake in maize under water-deficient conditions, revealing microbial contributions to stress tolerance trait [61,93]. Furthermore, insights from meta-analyses on rhizosphere-associated microbes in rice have demonstrated their role in enhancing resistance to environmental stressors, such as suboptimal soil pH and heavy metal toxicity [94]. These findings underscore the critical role of microbial associations in regulating plant stress responses, shaping trait selection criteria for breeding robust crop varieties. These diverse applications of RNA-Seq meta-analyses in addressing abiotic and biotic stress responses across various plant species are summarized in Table 4. To illustrate the diverse applications of RNA-seq meta-analysis in plant genomics, Fig. 2 provides a schematic representation.

This aggregated approach helps identify genes with stable expression changes that are crucial for stress adaptation, even if they are not consistently significant in individual studies. Furthermore, RNA-Seq meta-analysis helps identify conserved gene expression patterns across different plant species and conditions. This cross-species comparison can reveal core genes and pathways that are universally involved in plant stress responses or development. For instance, meta-analysis of RNA-Seq data from various plant species under drought stress has identified conserved genes and regulatory networks crucial for drought tolerance [59]. These conserved genes are likely to play fundamental roles in stress adaptation, making them valuable targets for further functional studies and genetic engineering. The integration of diverse data sources also helps uncover novel genes and pathways that might not be evident in single-study analyses. By pooling data from different experimental setups and conditions, meta-analysis can highlight genes that are consistently involved in important biological processes but might have been overlooked in isolated studies [59]. For example, meta-analysis of

Table 4 Studies utilizing meta-analysis to investigate plant stress responses.			
Stress Type (Biotic or Abiotic)	Stress	Species	Reference
Abiotic	Abscisic acid-related abiotic stresses	<i>Arabidopsis thaliana</i>	[2]
Abiotic	Hypoxia	<i>Arabidopsis</i> , rice	[56]
Abiotic	Salt stress	<i>Dunaliella</i> species	[4]
Biotic	Biotic stresses (fungal, viral, bacterial)	<i>Malus x domestica</i>	[102]
Abiotic	Drought responses	Various crops	[57]
Abiotic	Water stress	<i>Arabidopsis</i>	[17]
Abiotic	Abiotic stress	<i>Oryza sativa</i> (rice)	[100]
Abiotic	Abiotic stress	Various plant species	[58]
Abiotic	Biotic and abiotic stresses	<i>Oryza sativa</i> (rice)	[6]
Abiotic	Drought and cold stress	<i>Arabidopsis thaliana</i>	[103]
Biotic & Abiotic	Biotic and abiotic stresses	<i>Solanum lycopersicum</i> (tomato)	[104]
Abiotic & Biotic	Oxidative stress, combined biotic and abiotic stresses	<i>Helianthus annuus</i> (sunflower)	[105]
Abiotic	Multiple abiotic stresses	Various plant species	[106]
Biotic & Abiotic	Immune receptor genes	<i>Arabidopsis thaliana</i>	[96]

RNA-Seq data from plant pathogen interactions has led to the discovery of new genes and pathways involved in immune responses and disease resistance [95]. This approach has also enabled the identification of genes pivotal in managing multiple abiotic and biotic stressors through the interaction of key regulatory networks. For instance, pathways influenced by abscisic acid (ABA) have been shown to mediate stress tolerance mechanisms in *Arabidopsis thaliana* [2], while hypoxic responses in both *Arabidopsis* and rice reveal how plants adapt to oxygen-limiting conditions [56]. Research on responses to salt stress by Panahi et al. [4] further illustrates how meta-analysis can uncover complex regulatory networks across diverse conditions.

The integration of RNA-Seq data from various studies allows researchers to identify critical pathways and genes involved in managing multiple stressors. As demonstrated by Benny et al. [57], meta-analysis of RNA-Seq data effectively identified key genes and their associated chromosomal regions linked to drought responses in leaves across different crops. The studies by Cohen et al. [6] and Yang et al. [96] reveal how both biotic and abiotic stresses impact core transcriptomic responses and immune receptor genes in rice and *Arabidopsis*. Such analyses not only underscore the interconnected nature of plant stress responses but also emphasize the necessity of a holistic approach to deciphering complex stress biology, integrating molecular, environmental, and microbial dimensions.

RNA-Seq meta-analysis also provides valuable insights into the gene regulatory networks involved in growth and development. By integrating data from studies focusing on different developmental stages, tissues, or growth conditions, researchers can identify genes and pathways that are critical for processes like cell differentiation, organ formation, and growth regulation. For example, meta-analysis has been used to study the gene networks involved in root and shoot development, uncovering key transcription factors and signaling pathways that orchestrate these processes [97,98]. Moreover, RNA seq meta-analysis successfully applied for the candidate genes involve in seed development of Soybean [99]. This comprehensive approach helps researchers understand how developmental programs are controlled at the transcriptomic level and identifies potential genetic targets for improving growth traits.

Through meta-analysis, researchers can map intricate gene

Applications of RNA-Seq Meta-Analysis in Plant Genomics

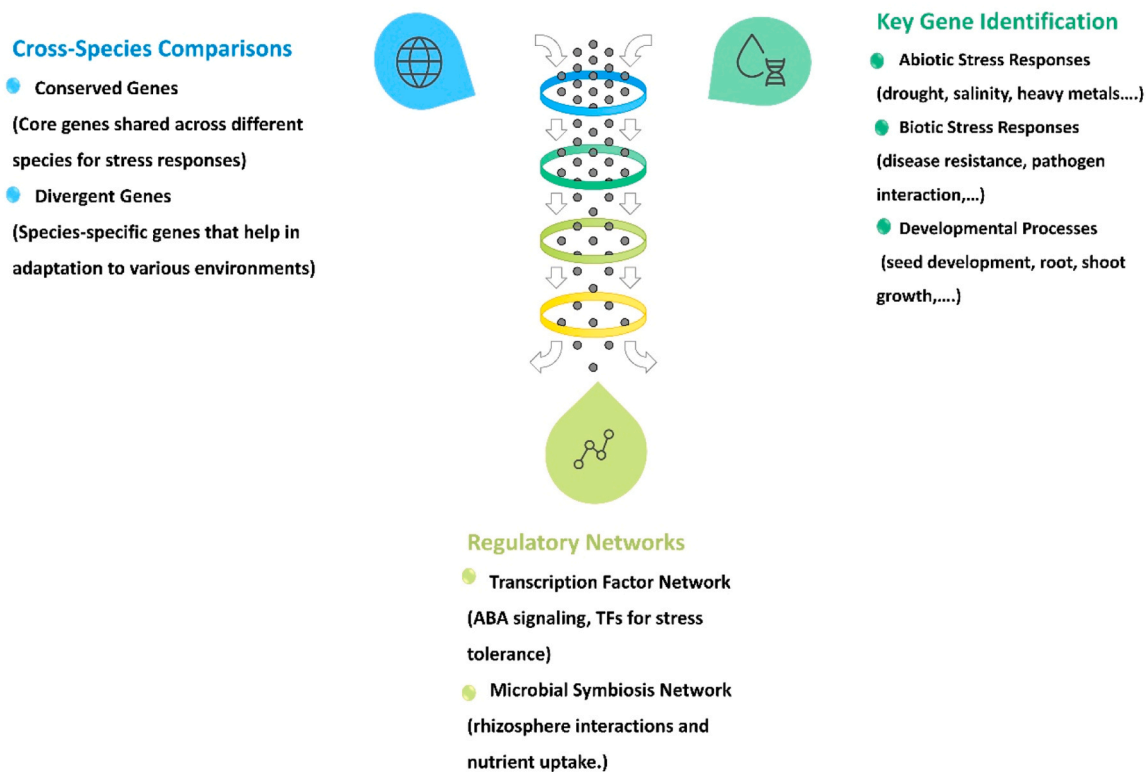


Fig. 2. Applications of RNA-Seq Meta-Analysis in Plant Genomics. This diagram illustrates the diverse applications of RNA-Seq meta-analysis in plant research.

regulatory networks by analyzing how different genes and their interactions are regulated across conditions and species. This approach helps identify upstream regulators (such as transcription factors and signaling molecules) and downstream target genes involved in complex biological processes. For example, the meta-analyses conducted by Buti et al. [100] and Balti et al. [58] on various plant species, respectively, reveal both conserved and species-specific responses to abiotic stresses, offering insights into how different plants cope with environmental challenges. Moreover, using a RNA S-seq meta-analysis, relation between the presence of TE superfamilies upstream, downstream or within introns of nearby genes and the differential expression of these genes in various stress conditions in the TE-poor *Arabidopsis thaliana* and the TE-rich *Solanum lycopersicum* were dissected by Deneweth et al. (2022) [101]. It was discovered a notable association between stress-responsive genes and nearby transposable elements (TEs) from various superfamilies. Moreover, it has been demonstrated that stress-specific TEs are predominantly located within introns and are more commonly found near up-regulated genes. Moreover, constructed regulatory networks of transcription factors (TFs), revealed how TE-mediated regulons are targeted by various TFs, including BRB/BPC, HD, HSF, GATA, NAC, DREB/CBF, and MYB factors in *Arabidopsis*, and AP2/ERF/B3, NAC, NF-Y, MYB, CXC, and HD factors in tomato [101]. To visually summarize these applications, we have included a schematic diagram (Fig. 2) that illustrates the primary research areas where RNA-Seq meta-analysis is applied in plant genomics. This diagram provides a clear overview of how RNA-Seq meta-analysis contributes to key gene identification, regulatory network prediction, and cross-species comparisons.

17.2. Exploration of evolutionary insights

RNA-Seq meta-analysis plays a pivotal role in understanding plant

evolution by providing insights into gene expression patterns, regulatory networks, and evolutionary processes across different species and conditions. RNA-Seq meta-analysis enables comparative transcriptomic studies across diverse plant species. By analyzing gene expression data from multiple studies, researchers can identify conserved and divergent gene expression patterns among closely related and distantly related plant species. This comparison helps to elucidate how gene expression has evolved and how different plant lineages have adapted to their environments. For instance, meta-analysis has been used to compare transcriptomic responses to environmental stresses across various plant species, revealing both conserved and lineage-specific adaptations [107, 108]. By comparing transcriptomic data from related species with different ecological niches or life strategies, researchers can trace how gene functions have diversified or specialized over evolutionary time. For example, meta-analysis has been used to study the evolution of stress-responsive genes in plants adapted to extreme environments, revealing how these genes have evolved to handle specific stressors [20, 57, 109]. This approach sheds light on how functional adaptations contribute to plant survival and success in varying ecological contexts.

Understanding the evolution of gene regulatory networks is another significant application of RNA-Seq meta-analysis. By analyzing gene expression data across different species and conditions, researchers can identify how regulatory interactions and pathways have evolved. RNA-Seq meta-analysis can uncover changes in transcription factor binding sites, gene interactions, and signaling pathways that have occurred during evolution. For instance, meta-analysis has revealed how changes in transcriptional regulatory networks have influenced the development of plant-specific traits such as flowering time, leaf morphology, and stress responses [110, 111]. These findings help in understanding the molecular basis of plant evolution and adaptation.

Moreover, by examining the consistency of gene expression across

different studies and species, researchers can determine which genes and pathways are fundamental to plant biology and which ones have evolved novel functions. For example, the meta-analysis of transcriptomic data has highlighted conserved pathways involved in basic cellular processes, while also uncovering lineage-specific innovations that contribute to plant diversity [112,113]. This dual approach helps in understanding the balance between evolutionary conservation and innovation.

17.3. Crop improvement and biotechnology

RNA-Seq meta-analysis significantly advances the development of plant genetic markers. By integrating transcriptomic data from various studies, researchers can identify genes that exhibit consistent differential expression under specific conditions or traits. These genes serve as valuable genetic markers for breeding programs. For example, by analyzing RNA-Seq data from plants under drought or disease stress, researchers can pinpoint genes that are reliably expressed in response to these stresses. These markers can then be utilized in marker-assisted selection (MAS) to breed plants with improved stress tolerance and disease resistance. This approach has been applied to identify markers linked to traits such as drought resistance in maize [114].

Moreover, RNA-Seq meta-analysis provides crucial insights into the genetic basis of yield-related traits. By analyzing transcriptomic data from various growth conditions and developmental stages, researchers can identify genes and regulatory networks that influence yield. This information guides engineering strategies to enhance crop productivity. For instance, genes involved in photosynthetic efficiency, nutrient uptake, and biomass accumulation can be targeted for genetic modification to boost yield [115]. Studies have shown that manipulating genes related to photosynthesis and nutrient utilization can significantly improve crop yield in species such as rice [116]. By understanding how these genes interact and regulate growth processes, researchers can develop crops with optimized yield potential.

RNA-Seq meta-analysis is a versatile approach to advancing genomic selection and breeding programs by identifying genes associated with desirable agronomic traits. In genomic selection, RNA-Seq meta-analysis aids in identifying genomic regions and candidate genes that are associated with specific traits. By integrating data from multiple RNA-Seq studies, researchers can perform genome-wide association studies (GWAS) that link gene expression profiles to agronomic traits. This information is used to develop genomic selection models that predict the performance of breeding lines based on their genetic makeup. For example, RNA-Seq meta-analysis has been employed to improve genomic selection models for traits such as drought tolerance in maize and disease resistance in wheat, leading to more accurate predictions and selection of superior breeding lines [117,118].

Moreover, RNA-Seq meta-analysis provides critical information for designing and optimizing breeding programs. By identifying genes and pathways associated with key agronomic traits, researchers can use this information to guide the selection of parent plants and breeding lines. For example, understanding the genetic basis of yield and disease resistance allows breeders to select parent plants that carry beneficial alleles for these traits. Additionally, meta-analysis can reveal gene interactions and regulatory networks that influence trait expression, enabling breeders to develop plants with improved performance through targeted genetic modifications or conventional breeding methods [119].

Benny et al. [120] conducted a meta-analysis of drought-related RNA-Seq datasets to identify key genes and its chromosome regions linked to drought responses in leaves across different crops. Several genes were proposed as biomarkers and breeding targets for drought tolerance. In maize, Jiang et al. [121] performed a meta-analysis to identify candidate genes of kernel row number trait. These genes were harnessed for developing improved cultivars of maize. These studies illustrate how RNA-Seq meta-analysis has evolved into a

hypothesis-generating and gene-prioritizing tool with significant implications for crop improvement.

17.4. Challenges and limitations

RNA-Seq meta-analysis in plants encounters several significant challenges that can impact the accuracy and reliability of the results. One primary challenge is dataset selection bias, which impacts the reliability and generalizability of results. Publicly available datasets exhibit considerable in experimental design, tissue type, developmental stage, environmental conditions, and sequencing depth. An uneven distribution of datasets—such as overrepresentation of certain genotypes, stress types, or treatment durations—can distort meta-analysis outcomes and neglect critical but underrepresented biological signals. Studies with small sample sizes or suboptimal quality may disproportionately influence aggregated findings if not properly filtered or weighted. Variations in normalization methods, batch effects, and metadata quality further complicate cross-study data integration. Therefore, careful curation of datasets, along with robust statistical frameworks that address inter-study heterogeneity, is essential to minimize bias and ensure meta-analysis results reflect biologically meaningful patterns rather than technical artifacts.

Data quality issues are another significant challenge in RNA-Seq meta-analysis. Inconsistent data quality across studies, including variability in sequencing depth, read length, and overall data quality, can impact the reliability of the meta-analysis results. High-quality, well-annotated data is crucial for accurate outcomes. Poor-quality sequencing data, low read counts, and incomplete annotations can skew results and affect the identification of differentially expressed genes [122]. Rigorous quality control measures and data filtering are necessary to ensure the integrity of the analysis and obtain reliable results. The statistical and computational demands of RNA-Seq meta-analysis are substantial. Managing and analyzing large-scale transcriptome data requires advanced statistical methods and significant computational resources. The complexity of integrating data from multiple studies, coupled with the need to account for various sources of variability, necessitates sophisticated analytical approaches. Statistical methods such as combining p-values or effect sizes must be robust to handle data heterogeneity and biases [123].

18. Conclusion and prospective

The integration of RNA-seq data through meta-analysis represents a significant advancement in plant genomics and biotechnology. This review examines the methodological landscape of RNA-seq meta-analysis, highlighting its potential to address dataset heterogeneity and yield broader insights into gene expression across various plant species and experimental conditions. By facilitating the detection of consistent expression patterns and regulatory mechanisms often overlooked in individual studies, meta-analysis enhances the reliability and interpretability of transcriptomic findings.

Our synthesis identifies several key contributions. Meta-analysis enables the identification of reproducible biomarkers and functional genes pertinent to plant breeding and stress adaptation. It improves gene function prediction and supports the development of crops with enhanced agronomic traits. However, challenges persist—particularly concerning the variability in experimental design, sequencing protocols, and preprocessing methods across studies. These discrepancies can limit dataset comparability and impact the accuracy of meta-analytical outcomes.

Standardization of RNA-seq workflows, including normalization strategies and quality control procedures, is crucial to minimizing these inconsistencies. Consistent data processing and annotation practices will enhance reproducibility and increase the reliability of downstream analyses. Furthermore, integrating transcriptomic meta-analysis with other omics layers—such as proteomics and metabolomics—can provide

a more comprehensive perspective on gene regulation and function. For example, the combination of transcriptomic and proteomic data has identified critical regulatory proteins involved in hypoxia responses in rice, illustrating the value of multi-omics integration.

Emerging technologies, particularly artificial intelligence (AI), present promising opportunities for advancing RNA-seq meta-analysis. Machine learning algorithms and predictive modeling tools can facilitate data integration, reveal complex patterns, and automate processes such as batch effect correction and trait selection. The HIVE platform, for instance, effectively integrates transcriptomic signals across multiple stress conditions, demonstrating AI's potential in agricultural genomics. AI also enhances the scalability of analyses and allows for adaptive refinement as new data become available—capabilities that are particularly advantageous in non-model species or large-scale comparative studies.

It is imperative that computational predictions are supported by experimental validation to confirm biological relevance. For instance, transcriptomic findings related to abscisic acid-mediated drought

tolerance in wheat have led to actionable insights for crop improvement. Expanding the application of meta-analysis across a broader range of crops, including non-model species, will deepen our understanding of both conserved and species-specific gene functions. This wider perspective is essential for enhancing crop resilience and advancing evolutionary and functional genomics.

In total, RNA-seq meta-analysis represents a transformative approach for integrating transcriptomic data across studies. When combined with AI-driven tools, rigorous standardization, and multi-omics strategies, it provides powerful insights into plant biology and accelerates progress in crop improvement. Continued innovation in analytical frameworks will be crucial to unlocking the full potential of RNA-seq data in both fundamental research and applied agricultural contexts.

Declaration of Competing Interest

The authors declare there is not any conflict of interest.

Appendix 1. - technical discussions on statistical models

Statistical Approaches for RNA-Seq Meta-Analysis

Modeling Methods

The fixed effect models calculated using following formula

$$\theta_{FE} = \frac{\sum_{i=1}^k w_i \theta_i}{\sum_{i=1}^k w_i}$$

Where:

- θ_{FE} is the fixed-effects estimate of the combined effect size.
- w_i is the weight of the i -th study, typically the inverse of the variance of the effect size estimate from that study
- θ_i is the effect size estimate from the i -th study.
- k is the number of studies

The weight w_i for each study is calculated as:

$$w_i = \frac{1}{Var(\theta_i)}$$

Where $Var(\theta_i)$ is the variance of the effect size estimate from the i -th study.

On the other hand, random-effects models, introduced in the latter half of the 20th century, address the limitations of fixed-effects models by accounting for variability across studies.

The formula for the combined effect size in a random-effects model is:

$$\theta_{RE} = \frac{\sum_{i=1}^k w_i \theta_i}{\sum_{i=1}^k w_i}$$

Where:

- θ_{RE} is the random effects estimate of the combined effect size
- w_i is the weight of the i -th study in the random-effect model which accounts for between-study variability
- θ_i is the effect size estimate from the i -th study
- k is the number of studies

$$w_i = \frac{1}{Var(\theta_i) + \tau^2}$$

Where:

- $Var(\theta_i)$ is the variance of the effect size estimate from the i -th study

- r^2 is the estimated between-study variance (also known as the random-effects

In the random-effects model, r^2 represents the variability of the true effect sizes across studies and is estimated using methods such as DerSimonian-Laird method or restricted maximum likelihood (REML).

The fixed-effects models calculated using following formula;

$$\theta_{FE} = \frac{\sum_{i=1}^k w_i \theta_i}{\sum_{i=1}^k w_i}$$

Where:

- θ_{FE} is the fixed-effects estimate of the combined effect size.
- w_i is the weight of the i -th study, typically the inverse of the variance of the effect size estimate from that study
- θ_i is the effect size estimate from the i -th study.
- k is the number of studies

The weight w_i for each study is calculated as:

$$w_i = \frac{1}{Var(\theta_i)}$$

Where $Var(\theta_i)$ is the variance of the effect size estimate from the i -th study.

Fisher's combined probability test is a widely used method for meta-analyzing p-values from different studies [1]. This test combines the p-values into a single test statistic, which is then used to assess overall significance. The formula for Fisher's method is as follows:

$$\chi^2 = -2 \sum_{i=1}^k \ln(p_i)$$

Where:

- p_i is the p-value from the i -th study,
- k is the number of studies

The test statistic χ^2 follows a chi-squared distribution with $2k$ degrees of freedom. Fisher's method is advantageous because it is straightforward and does not rely on assumptions about the distribution of effect sizes across studies. However, it is sensitive to the number of studies and can be influenced by extreme p-values[2].

The formula for Stouffer's method is:

$$Z_{combined} = \frac{\sum_{i=1}^k z_i}{\sqrt{\sum_{i=1}^k w_i}}$$

Where:

- z_i is the Z-score corresponding to the p-value from the i -th study,
- w_i is the weight for the i -th study, often based on sample size or study precision.

1.Hu Z, Wu Y, Zhou Y: **Meta-analysis of RNA-seq studies with an adaptive weight and truncation p-value combination test.** *Applied Mathematical Modeling* 2024.

2.Van Aert RC, Wicherts JM, van Assen MA: **Conducting meta-analyses based on p values: Reservations and recommendations for applying p-uniform and p-curve.** *Perspectives on Psychological Science* 2016, 11(5):713–729.

Data availability

Data will be made available on request.

References

- [1] M. Shintani, H. Bono, Meta-analysis of public RNA-sequencing data of drought and salt stresses in different phenotypes of *Oryza sativa*, *bioRxiv* (2024) 605779, 2024.2008. 2006.
- [2] M. Shintani, K. Tamura, H. Bono, Meta-analysis of public RNA sequencing data of abscisic acid-related abiotic stresses in *Arabidopsis thaliana*, *Front. Plant Sci.* 15 (2024) 1343787.
- [3] G. Calia, S. Marguerit, A.P. Zotta Mota, M. Vidal, M. Seynabou-Fall, H.T. Nguyen, A. Bhat, H. Schuler, C. Gwizdek, A.C.M. Brasileiro, Disentangling plant response to biotic and abiotic stress using HIVE, a novel tool to perform unpaired multi-transcriptomics integration, 2024:2024.2003, *bioRxiv* (2004) 583290.
- [4] B. Panahi, M. Frahadian, J.T. Dums, M.A. Hejazi, Integration of cross species RNA-Seq meta-analysis and machine-learning models identifies the most important salt stress-responsive pathways in microalga *Dunaliella*, *Front. Genet.* 10 (2019) 752.
- [5] P. Tyagi, D. Singh, S. Mathur, A. Singh, R. Ranjan, Upcoming progress of transcriptomics studies on plants: an overview, *Front. Plant Sci.* 13 (2022) 1030890.
- [6] S.P. Cohen, J.E. Leach, Abiotic and biotic stresses induce a core transcriptome response in rice, *Sci. Rep.* 9 (1) (2019) 6273.
- [7] A. Rau, G. Marot, F. Jaffrézic, Differential meta-analysis of RNA-seq data from multiple studies, *BMC Bioinforma.* 15 (2014) 1–10.
- [8] M. Jain, Next-generation sequencing technologies for gene expression profiling in plants, *Brief. Funct. Genom.* 11 (1) (2012) 63–70.

- [9] X. Zhang, Z. Ibrahim, M.B. Khaskheli, H. Raza, F. Zhou, I.H. Shamsi, Integrative approaches to abiotic stress management in crops: combining bioinformatics educational tools and artificial intelligence applications, *Sustainability* 16 (17) (2024) 7651.
- [10] S. Kamali, A. Singh, Genomic and transcriptomic approaches to developing abiotic stress-resilient crops, *Agronomy* 13 (12) (2023) 2903.
- [11] B.N. Keel, A.K. Lindholm-Perry, Recent developments and future directions in meta-analysis of differential gene expression in livestock RNA-Seq, *Front. Genet.* 13 (2022) 983043.
- [12] B. Panahi, M. Farhadian, N.H. Gharajeh, S.A. Mohammadi, M.A. Hejazi, Meta-analysis of transcriptomic profiles in *Dunaliella tertiolecta* reveals molecular pathway responses to different abiotic stresses, *Funct. Plant Biol.* 51 (3) (2024).
- [13] A.M. Eteleeb, H.N. Moseley, E.C. Rouchka, A comparison of combined p-value methods for gene differential expression using RNA-seq data, *Proc. 5th ACM Conf. Bioinforma. Comput. Biol. Health Inform.* 2014 (2014) 417–425.
- [14] H.T.N. Tran, K.S. Ang, M. Chevrier, X. Zhang, N.Y.S. Lee, M. Goh, J. Chen, A benchmark of batch-effect correction methods for single-cell RNA sequencing data, *Genome Biol.* 21 (2020) 1–32.
- [15] D. Lakens, J. Hilgard, J. Staaks, On the reproducibility of meta-analyses: six practical recommendations, *BMC Psychol.* 4 (2016) 1–10.
- [16] A. Negi, A. Shukla, A. Jaiswar, J. Shrinet, R.S. Jasrotia, Applications and challenges of microarray and RNA-sequencing, *Bioinformatics* (2022) 91–103.
- [17] J.S. Rest, O. Wilkins, W. Yuan, M.D. Purugganan, J. Gurevitch, Meta-analysis and meta-regression of transcriptomic responses to water stress in *Arabidopsis*, *Plant J.* 85 (4) (2016) 548–560.
- [18] A. Rau, G. Marot, F. Jaffrézic, Differential meta-analysis of RNA-seq data from multiple studies, *BMC Bioinforma.* 15 (1) (2014) 91.
- [19] Y. Han, S. Gao, K. Muegge, W. Zhang, B. Zhou, Advanced applications of RNA sequencing and challenges, *Bioinforma. Biol. Insights* 9 (2015) S28991 (BBI).
- [20] Y. Wang, T. Li, Z. Sun, X. Huang, N. Yu, H. Tai, Q. Yang, Comparative transcriptome meta-analysis reveals a set of genes involved in the responses to multiple pathogens in maize, *Front. Plant Sci.* 13 (2022) 971371.
- [21] M.N. Saidi, I. Mahjoubi, I. Yacoubi, Transcriptome meta-analysis of abiotic stresses-responsive genes and identification of candidate transcription factors for broad stress tolerance in wheat, *Protoplasma* 260 (3) (2023) 707–721.
- [22] N. Rawat, S.P. Kiran, D. Du, F.G. Gmitter Jr, Z. Deng, Comprehensive meta-analysis, co-expression, and miRNA nested network analysis identifies gene candidates in citrus against Huanglongbing disease, *BMC Plant Biol.* 15 (1) (2015) 184.
- [23] J. Costa-Silva, D.S. Domingues, D. Menotti, M. Hungria, F.M. Lopes, Temporal progress of gene expression analysis with RNA-Seq data: a review on the relationship between computational methods, *Comput. Struct. Biotechnol. J.* 21 (2023) 86–98.
- [24] D. Deshpande, K. Chhugani, Y. Chang, A. Karlsberg, C. Loeffler, J. Zhang, A. Muszyńska, V. Munteanu, H. Yang, J. Rotman, RNA-seq data science: From raw data to effective interpretation, *Front. Genet.* 14 (2023) 997383.
- [25] S. Sircar, N. Parekh, Meta-analysis of drought-tolerant genotypes in *Oryza sativa*: a network-based approach, *PLOS One* 14 (5) (2019) e0216068.
- [26] L. Hagverd, A.T. Lun, M.D. Morgan, J.C. Marioni, Batch effects in single-cell RNA-sequencing data are corrected by matching mutual nearest neighbors, *Nat. Biotechnol.* 36 (5) (2018) 421–427.
- [27] N. Ghahramani, J. Shodja, S.A. Rafat, B. Panahi, K. Hasanpur, Integrative systems biology analysis elucidates mastitis disease underlying functional modules in dairy cattle, *Front. Genet.* 12 (2021) 712306.
- [28] M.I. Love, W. Huber, S. Anders, Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2, *Genome Biol.* 15 (12) (2014) 550.
- [29] M. Smid, R.R. Coebergh van den Braak, H.J. van de Werken, J. van Riet, A. van Galen, V. de Weerd, M. van der Vlugt-Daane, S.I. Bril, Z.S. Lalmahomed, W. P. Kloosterman, Gene length corrected trimmed mean of M-values (GeTMM) processing of RNA-seq data performs similarly in intersample analyses while improving intrasample comparisons, *BMC Bioinforma.* 19 (1) (2018) 236.
- [30] Y. Zhao, L. Wong, W.W.B. Goh, How to do quantile normalization correctly for gene expression data analyses, *Sci. Rep.* 10 (1) (2020) 15534.
- [31] R.D. Tihagam, S. Bhatnagar, A multi-platform normalization method for meta-analysis of gene expression data, *Methods* 217 (2023) 43–48.
- [32] F. Del Carratore, A. Jankevics, R. Eisinga, T. Heskes, F. Hong, R. Breitling, RankProd 2.0: a refactored bioconductor package for detecting differentially expressed features in molecular profiling datasets, *Bioinformatics* 33 (17) (2017) 2774–2775.
- [33] S.M. Lin, P. Du, W. Huber, W.A. Kibbe, Model-based variance-stabilizing transformation for Illumina microarray data, *Nucleic Acids Res.* 36 (2) (2008) e11 e11.
- [34] P.I. Kontou, A. Pavlopoulou, P.G. Bagos, Methods of analysis and meta-analysis for identifying differentially expressed genes, *Genetic Epidemiology: Methods and Protocols*, Springer, 2018, pp. 183–210.
- [35] J.E. Hunter, F.L. Schmidt, Fixed effects vs. random effects meta-analysis models: Implications for cumulative research knowledge, *Int. J. Sel. Assess.* 8 (4) (2000) 275–292.
- [36] M. Borenstein, L. Hedges, H. Rothstein, *Meta-analysis: Fixed effect vs. random effects*, *Meta-analysis com* 2007, pp. 1–162..
- [37] C. Jia, Y. Hu, Y. Liu, M. Li, Mapping splicing quantitative trait loci in RNA-Seq: supplementary issue: sequencing platform modeling and analysis, *Cancer Inform.* 14 (2015) S24832 (CIN).
- [38] Z. Cai, X. Fan, A comparison of fixed-effects and random-effects models for multivariate meta-analysis using an SEM approach, *Multivar. Behav. Res.* 55 (6) (2020) 839–854.
- [39] S.F. Anderson, K. Kelley, Sample size planning for replication studies: the devil is in the design, *Psychol. Methods* (2022).
- [40] A. Guolo, C. Varin, Random-effects meta-analysis: the number of studies matters, *Stat. Methods Med. Res.* 26 (3) (2017) 1500–1518.
- [41] U. Siangphoe, K.J. Archer, Estimation of random effects and identifying heterogeneous genes in meta-analysis of gene expression studies, *Brief. Bioinforma.* 18 (4) (2017) 602–618.
- [42] M. Breuer, E. Dehaan, Using and interpreting fixed effects models, *J. Account. Res.* 62 (4) (2024) 1183–1226.
- [43] Z. Hu, Y. Wu, Y. Zhou, Meta-analysis of RNA-seq studies with an adaptive weight and truncation p-value combination test, *Appl. Math. Model.* (2024).
- [44] B.S. Pedersen, D.A. Schwartz, I.V. Yang, K.J. Kechris, Comb-p: software for combining, analyzing, grouping and correcting spatially correlated P-values, *Bioinformatics* 28 (22) (2012) 2986–2988.
- [45] G. Marot, F. Jaffrézic, A. Raunet, *RNASeq: Differential meta-analysis of RNA-seq data*, *dim (param)* 2020, 1(26408):3..
- [46] D. Fanidis, P. Moulos, Integrative, normalization-insusceptible statistical analysis of RNA-Seq data, with improved differential expression and unbiased downstream functional analysis, *Brief. Bioinforma.* 22 (3) (2021) bbaa156.
- [47] J.A. Villatoro-García, J. Martorell-Marugán, D. Toro-Domínguez, Y. Román-Montoya, P. Femia, P. Carmona-Sáez, DEXMA: An R package for performing gene expression meta-analysis with missing genes, *Mathematics* 10 (18) (2022) 3376.
- [48] K. Tsuyuzaki, I. Nikaido, *metaSeq: Meta-Analysis of RNA-Seq Count Data*, Tokyo University of Science, Tokyo, 2013.
- [49] D.A. Forero, Available software for meta-analyses of genome-wide expression studies, *Curr. Genom.* 20 (5) (2019) 325–331.
- [50] Wang X., Li J., Tseng G.C., Wang M.X.: Package ‘MetaDE’. 2012.
- [51] X. Xing, J.S. Liu, W. Zhong, MetaGen: reference-free learning with multiple metagenomic samples, *Genome Biol.* 18 (2017) 1–15.
- [52] Y. Du, X. Zhen, L. Shao, C.G. Snoek, Metanorm: learning to normalize few-shot batches across domains, *Int. Conf. Learn. Represent.* 2020 (2020).
- [53] Viechtbauer W., Viechtbauer M.W.: Package ‘metafor’. The Comprehensive R Archive Network Package ‘metafor’ (<http://cran.r-project.org/web/packages/metafor/metafor.pdf> 2015.
- [54] S. Blanck, G. Marot, SMAGEXP: a galaxy tool suite for transcriptomics data meta-analysis, *GigaScience* 8 (2) (2019) giy167.
- [55] R. Barker, C.P. Kruse, C. Johnson, A. Saravia-Butler, H. Fogle, H.-S. Chang, R. M. Trane, N. Kinscherf, A. Villacampa, A. Manzano, Meta-analysis of the space flight and microgravity response of the *Arabidopsis* plant transcriptome, *npj Microgravity* 9 (1) (2023) 21.
- [56] K. Tamura, H. Bono, Meta-analysis of RNA sequencing data of *Arabidopsis* and rice under hypoxia, *Life* 12 (7) (2022) 1079.
- [57] J. Benny, A. Pisciotto, T. Caruso, F. Martinelli, Identification of key genes and its chromosome regions linked to drought responses in leaves across different crops through meta-analysis of RNA-Seq data, *BMC Plant Biol.* 19 (2019) 1–18.
- [58] I. Balti, J. Benny, A. Perrone, T. Caruso, D. Abdallah, A. Salhi-Hannachi, F. Martinelli, Identification of conserved genes linked to responses to abiotic stresses in leaves among different plant species, *Funct. Plant Biol.* 48 (1) (2020) 54–71.
- [59] L. Shaar-Moshe, S. Hübner, Z. Peleg, Identification of conserved drought-adaptive genes using a cross-species meta-analysis approach, *BMC Plant Biol.* 15 (2015) 1–18.
- [60] Z. Hu, Y. Wu, Y. Zhou, Meta-analysis of RNA-seq studies with an adaptive weight and truncation p-value combination test, *Appl. Math. Model.* 136 (2024) 115611.
- [61] M. Miransari, Contribution of arbuscular mycorrhizal symbiosis to plant growth under different types of soil stress, *Plant Biol.* 12 (4) (2010) 563–569.
- [62] S. Nakagawa, I.C. Cuthill, Effect size, confidence interval and statistical significance: a practical guide for biologists, *Biol. Rev.* 82 (4) (2007) 591–605.
- [63] H.E. Augusteijn, R. van Aert, M.A. van Assen, The effect of publication bias on the Q test and assessment of heterogeneity, *Psychol. Methods* 24 (1) (2019) 116.
- [64] J.P. Higgins, S.G. Thompson, Quantifying heterogeneity in a meta-analysis, *Stat. Med.* 21 (11) (2002) 1539–1558.
- [65] M.A. Van Assen, R. van Aert, J.M. Wicherts, Meta-analysis using effect size distributions of only statistically significant studies, *Psychol. Methods* 20 (3) (2015) 293.
- [66] O. Irshad, M.U. Khan, A comparative analysis of biological data integration systems famous for data exploitation and knowledge discovery, *Curr. Bioinforma.* 16 (5) (2021) 662–681.
- [67] B. Panahi, S. Golkari, Identification of gene expression signature for drought stress response in barley (*Hordeum vulgare* L.) using machine learning approach, *Curr. Plant Biol.* 39 (2024) 100370.
- [68] B. Panahi, A. Shahi, Transcriptome data mining in combination with co-expression network analysis identifies the functional modules and critical regulators in *Hordeum vulgare* L. in response to cold stress, *Biochem. Biophys. Rep.* 37 (2024) 101620.
- [69] M. Egger, G.D. Smith, M. Schneider, C. Minder, Bias in meta-analysis detected by a simple, graphical test, *bmj* 315 (7109) (1997) 629–634.
- [70] W. Viechtbauer, Conducting meta-analyses in R with the metafor package, *J. Stat. Softw.* 36 (3) (2010) 1–48.
- [71] W. Viechtbauer, M.W. Viechtbauer, Package ‘metafor’. the comprehensive R archive network. Package ‘metafor’, *J. Stat. Softw.* 36 (2015) 1–48.
- [72] H. Kazemi, A. Sabouri, A. Aalami, A. Abedi, A comprehensive meta-analysis to identify the responsive genes in sorghum under salinity and drought stresses (*Sorghum bicolor*), *J. Plant Growth Regul.* 42 (11) (2023) 7096–7115.

- [73] K. Athanasopoulou, V.-I. Michalopoulou, A. Scorilas, P.G. Adamopoulos, Integrating artificial intelligence in next-generation sequencing: advances, challenges, and future directions, *Curr. Issues Mol. Biol.* 47 (6) (2025) 470.
- [74] B. Panahi, R. Hamid, H.M.Z. Jalaly, Deciphering plant transcriptomes: leveraging machine learning for deeper insights, *Curr. Plant Biol.* (2024) 100432.
- [75] S. Babichev, I. Liakh, I. Kalinina, Applying a recurrent neural network-based deep learning model for gene expression data classification, *Appl. Sci.* 13 (21) (2023) 11823.
- [76] M. Daoud, Autoencoder-Based Techniques for Improved Classification in Settings with High Dimensional and Small Sized Data, The University of Waikato, 2020.
- [77] L. Su, C. Xu, S. Zeng, L. Su, T. Joshi, G. Stacey, D. Xu, Large-scale integrative analysis of soybean transcriptome using an unsupervised autoencoder model, *Front. Plant Sci.* 13 (2022) 831204.
- [78] G. Calia, S. Marguerit, A.P.Z. Mota, M. Vidal, M. Seynabou-Fall, H.T. Nguyen, A. Bhat, H. Schuler, C. Gwizdek, A.C.M. Brasileiro, Disentangling plant response to biotic and abiotic stress using HIVE, a novel tool to perform unpaired multi-transcriptomics integration, *BioRxiv* (2024).
- [79] J. Yan, X. Wang, Unsupervised and semi-supervised learning: the next frontier in machine learning for plant systems biology, *Plant J.* 111 (6) (2022) 1527–1538.
- [80] B.M. Moore, P. Wang, P. Fan, A. Lee, B. Leong, Y.-R. Lou, C.A. Schenck, K. Sugimoto, R. Last, M.D. Lehti-Shiu, Within-and cross-species predictions of plant specialized metabolism genes using transfer learning, *Silico Plants* 2 (1) (2020) diaa005.
- [81] E.K. Innocent, Enhancing Data Security in Healthcare with Synthetic Data Generation: An Autoencoder and Variational Autoencoder Approach, Oslo Metropolitan University, 2024.
- [82] H.T. Otal, A. Subasi, F. Kurt, M.A. Canbaz, Y. Uzun, Analysis of gene regulatory networks from gene expression using graph neural networks. *Digital Healthcare, Digital Transformation and Citizen Empowerment in Asia-Pacific and Europe for a Healthier Society*, Elsevier, 2025, pp. 249–270.
- [83] K. Asefpoor Vakilian, Machine learning improves our knowledge about miRNA functions towards plant abiotic stresses, *Sci. Rep.* 10 (1) (2020) 3041.
- [84] P.H. Sudmant, M.S. Alexis, C.B. Burge, Meta-analysis of RNA-seq expression data across species, tissues and studies, *Genome Biol.* 16 (1) (2015) 287.
- [85] R. Shamloo-Dashtpajardi, M.J. Tanin, M. Aliakbari, D.K. Saini, Unveiling the role of the ERD15 gene in wheat's tolerance to combined drought and salinity stress: a meta-analysis of QTL and RNA-Seq data, *Physiol. Plant.* 176 (5) (2024) e14570.
- [86] M. Rodriguez, A. Scintu, C.M. Posadinu, Y. Xu, C.V. Nguyen, H. Sun, E. Bitocchi, E. Bellucci, R. Papa, Z. Fei, GWAS based on RNA-Seq SNPs and high-throughput phenotyping combined with climatic data highlights the reservoir of valuable genetic diversity in regional tomato landraces, *Genes* 11 (11) (2020) 1387.
- [87] E. Gayathiri, D. Viswanathan, P. Prakash, S. Pandiaraj, B.K. Kondapavuluri, C. D. Singh, R. Thiruvengadam, S. Balu, R. Anantharaman, A. Gaur, A critical review on epigenetics and epigenomics in plant development and stress resilience, *Braz. J. Bot.* 48 (1) (2025) 51.
- [88] G.P. Saradadevi, D. Das, S.K. Mangrauthia, S. Mohapatra, C. Chikkaputtaiah, M. Roorkiwal, M. Solanki, R.M. Sundaram, N.N. Chirravuri, A.S. Sakhare, Genetic, epigenetic, genomic and microbial approaches to enhance salt tolerance of plants: a comprehensive review, *Biology* 10 (12) (2021) 1255.
- [89] N. Wang, S. Feng, X. Ma, Q. Chen, C. Liu, Z. Qi, Meta-analysis and multiomics of a chromosome segment substitution line reveal candidate genes associated with seed hardness in soybean, *J. Agric. Food Chem.* 71 (44) (2023) 16840–16854.
- [90] P. Manchanda, P. Chaudhary, R. Deswal, Photosynthesis regulation, cell membrane stabilization and methylglyoxal detoxification seems major altered pathways under cold stress as revealed by integrated multi-omics meta-analysis, *Physiol. Mol. Biol. Plants* 29 (10) (2023) 1395–1407.
- [91] A.E. Melton, S.J. Galla, C.D.C. Dumaguit, J.M. Wajohn, S. Novak, M. Serpe, P. Martinez, S. Buerki, Meta-analysis reveals challenges and gaps for genome-to-phenome research underpinning plant drought response, *Int. J. Mol. Sci.* 23 (20) (2022) 12297.
- [92] Μακρόπουλος KA: Meta-analysis of large-scale RNA sequencing (RNA-seq) data to study the effect of abiotic environmental variables on rice (*Oryza sativa*). 2023.
- [93] A.M.M. Silva, Arbuscular Mycorrhizal Fungi and Rhizobacteria Attenuating Drought Stress in Maize Plants, Universidade de São Paulo, 2023.
- [94] H. Etesami, G.A. Beattie, Plant-microbe interactions in adaptation of agricultural crops to abiotic stress conditions, *Probiot. Plant Health* (2017) 163–200.
- [95] Y. Biniiaz, A. Tahmasebi, A. Tahmasebi, B.R. Albrechtsen, P. Pocai, A. Afsharifar, Transcriptome meta-analysis identifies candidate hub genes and pathways of pathogen stress responses in *Arabidopsis thaliana*, *Biology* 11 (8) (2022) 1155.
- [96] L. Yang, Z. Wang, J. Hua, A meta-analysis reveals opposite effects of biotic and abiotic stresses on transcript levels of *Arabidopsis* intracellular immune receptor genes, *Front. Plant Sci.* 12 (2021) 625729.
- [97] N.V. Hoang, C. Park, M. Kamran, J.-Y. Lee, Gene regulatory network guided investigations and engineering of storage root development in root crops, *Front. Plant Sci.* 11 (2020) 762.
- [98] B.A. López-Rufz, B. García-Ponce, M. de la Paz Sánchez, E.R. Álvarez-Buylla, A. O. Urrutia, A. Garay-Arroyo, Genome-wide association studies meta-analysis uncovers NOJO and SGS3 novel genes involved in *Arabidopsis thaliana* primary root development and plasticity, *Mol. Biol. Rep.* 51 (1) (2024) 763.
- [99] Z. Qi, Z. Zhang, Z. Wang, J. Yu, H. Qin, X. Mao, H. Jiang, D. Xin, Z. Yin, R. Zhu, Meta-analysis and transcriptome profiling reveal hub genes for soybean seed storage composition during seed development, *Plant Cell Environ.* 41 (9) (2018) 2109–2127.
- [100] M. Buti, E. Baldoni, E. Formentin, J. Milc, G. Frugis, F. Lo Schiavo, A. Genga, E. Francia, A meta-analysis of comparative transcriptomic data reveals a set of key genes involved in the tolerance to abiotic stresses in rice, *Int. J. Mol. Sci.* 20 (22) (2019) 5662.
- [101] J. Deneweth, Y. Van de Peer, V. Vermeirssen, Nearby transposable elements impact plant stress gene regulatory networks: a meta-analysis in *A. thaliana* and *S. lycopersicum*, *BMC Genom.* 23 (1) (2022) 18.
- [102] B. Balan, F.P. Marra, T. Caruso, F. Martinelli, Transcriptomic responses to biotic stresses in *Malus x domestica*: a meta-analysis study, *Sci. Rep.* 8 (1) (2018) 1970.
- [103] R. Sharma, G. Singh, S. Bhattacharya, A. Singh, Comparative transcriptome meta-analysis of *Arabidopsis thaliana* under drought and cold stress, *PLOS One* 13 (9) (2018) e0203266.
- [104] E. Ashrafi-Dehkordi, A. Alemzadeh, N. Tanaka, H. Razi, Meta-analysis of transcriptomic responses to biotic and abiotic stress in tomato, *PeerJ* 6 (2018) e4631.
- [105] V.S. Ramu, A. Paramanatham, V. Ramegowda, B. Mohan-Raju, M. Udayakumar, M. Senthil-Kumar, Transcriptome analysis of sunflower genotypes with contrasting oxidative stress tolerance reveals individual-and combined-biotic and abiotic stress tolerance mechanisms, *PLOS One* 11 (6) (2016) e0157522.
- [106] R. Sanchez-Munoz, T. Depaepe, M. Samalova, J. Hejatk, I. Zaplana, D. Van Der Straeten, The molecular core of transcriptome responses to abiotic stress in plants: a machine learning-driven meta-analysis, *bioRxiv* (2024) 576978, 2024.2001.
- [107] T.M. Winkelmüller, Comparative Transcriptomics within *Arabidopsis thaliana* Accessions and Across Brassicaceae Species Reveal Evolutionary Conserved and Lineage-specific Expression Signatures in Pattern-triggered Immunity, *Universität zu Köln*, 2018.
- [108] W. Fan, C. Liu, B. Cao, S. Ma, J. Hu, Z. Xiang, A. Zhao, A meta-analysis of transcriptomic profiles reveals molecular pathways response to cadmium stress of Gramineae, *Ecotoxicol. Environ. Saf.* 209 (2021) 111816.
- [109] E. Baldoni, G. Frugis, F. Martinelli, J. Benny, D. Paffetti, M. Buti, A comparative transcriptomic meta-analysis revealed conserved key genes and regulatory networks involved in drought tolerance in cereal crops, *Int. J. Mol. Sci.* 22 (23) (2021) 13062.
- [110] J. Kuppler, M.M. Kotowska, A meta-analysis of responses in floral traits and flower-visitor interactions to water deficit, *Glob. Change Biol.* 27 (13) (2021) 3095–3108.
- [111] M. Canton, C. Forestan, C. Bonghi, S. Varotto, Meta-analysis of RNA-Seq studies reveals genes with dominant functions during flower bud endo-to eco-dormancy transition in *Prunus* species, *Sci. Rep.* 11 (1) (2021) 13173.
- [112] P.H. Sudmant, M.S. Alexis, C.B. Burge, Meta-analysis of RNA-seq expression data across species, tissues and studies, *Genome Biol.* 16 (2015) 1–11.
- [113] Y. Sheng, R.A. Ali, A. Heyland, Comparative transcriptomics analysis pipeline for the meta-analysis of phylogenetically divergent datasets (CoRMAP), *BMC Bioinforma.* 23 (1) (2022) 415.
- [114] G. Pal, R. Bakade, S. Deshpande, V. Sureshkumar, S.S. Patil, A. Dawane, S. Agarwal, V. Niranjani, M. Prasanna Kumar, R.S. Vemanna, Transcriptomic responses under combined bacterial blight and drought stress in rice reveal potential genes to improve multi-stress tolerance, *BMC Plant Biol.* 22 (1) (2022) 349.
- [115] F. Loni, A. Ismaili, B. Nakhoda, H. Darzi Ramandi, Z.-S. Shobbar, The genomic regions and candidate genes associated with drought tolerance and yield-related traits in foxtail millet: an integrative meta-analysis approach, *Plant Growth Regul.* 101 (1) (2023) 169–185.
- [116] P. Daryani, N. Amirbakhtiar, J. Soorni, F. Loni, H. Darzi Ramandi, Z.-S. Shobbar, Uncovering the genomic regions associated with yield maintenance in rice under drought stress using an integrated meta-analysis approach, *Rice* 17 (1) (2024) 7.
- [117] R. Mirdar Mansuri, Z.-S. Shobbar, N. Babaeian Jelodar, M. Ghaffari, S. M. Mohammadi, P. Daryani, Salt tolerance involved candidate genes in rice: an integrative meta-analysis approach, *BMC Plant Biol.* 20 (2020) 1–14.
- [118] X. Zhao, Y. Niu, Z. Hossain, J. Shi, T. Mao, X. Bai, Integrated QTL mapping, meta-analysis, and RNA-sequencing reveal candidate genes for maize deep-sowing tolerance, *Int. J. Mol. Sci.* 24 (7) (2023) 6770.
- [119] B.K. Singh, S. Venkadesan, M. Ramkumar, P. Shanmugavadeivel, B. Dutta, C. Prakash, M. Pal, A.U. Solanke, A. Rai, N.K. Singh, Meta-analysis of microarray data and their utility in dissecting the mapped QTLs for heat acclimation in rice, *Plants* 12 (8) (2023) 1697.
- [120] J. Benny, A. Pisciotto, T. Caruso, F. Martinelli, Identification of key genes and its chromosome regions linked to drought responses in leaves across different crops through meta-analysis of RNA-Seq data, *BMC Plant Biol.* 19 (1) (2019) 194.
- [121] Q. Jiang, D. Tang, C. Hu, J. Qu, J. Liu, Combining meta-QTL with RNA-seq data to identify candidate genes of kernel row number trait in maize, *Maydica* 61 (4) (2016) 1–9.
- [122] H.A. Chowdhury, D.K. Bhattacharyya, J.K. Kalita, Differential expression analysis of RNA-seq reads: overview, taxonomy, and tools, *IEEE/ACM Trans. Comput. Biol. Bioinforma.* 17 (2) (2018) 566–586.
- [123] R.B. Darlington, A.F. Hayes, Combining independent p values: extensions of the Stouffer and binomial methods, *Psychol. Methods* 5 (4) (2000) 496.